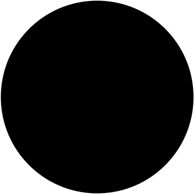
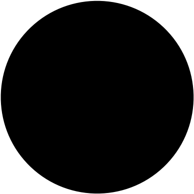
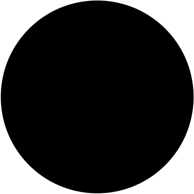
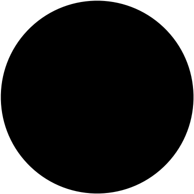
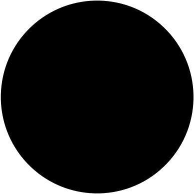
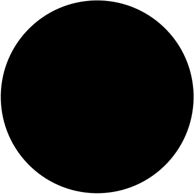


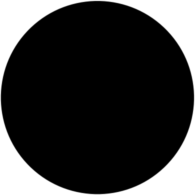












11 x_1^{l-1} f 1

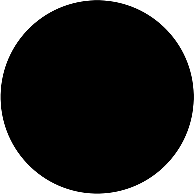
1

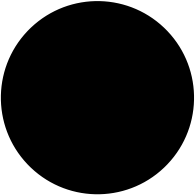
11 x_1^{l} f 1

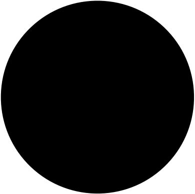
1

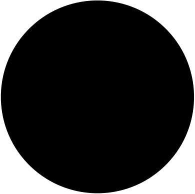
2

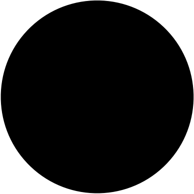
3



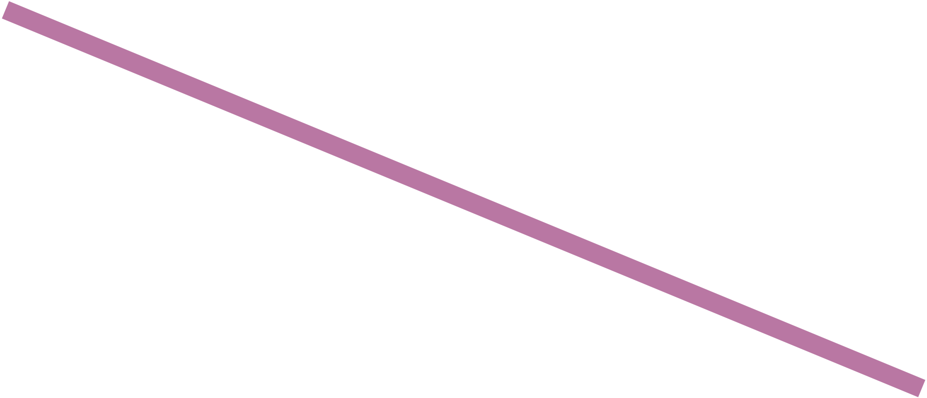


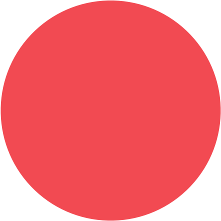


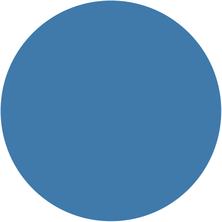


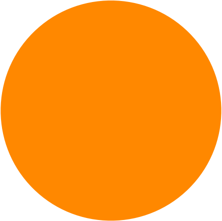


4









11 x_1^{l-1} f 1

22 x_2^l f 2

12

12 x_1^{l-1} f 2

1234

34 x_3^l f 4

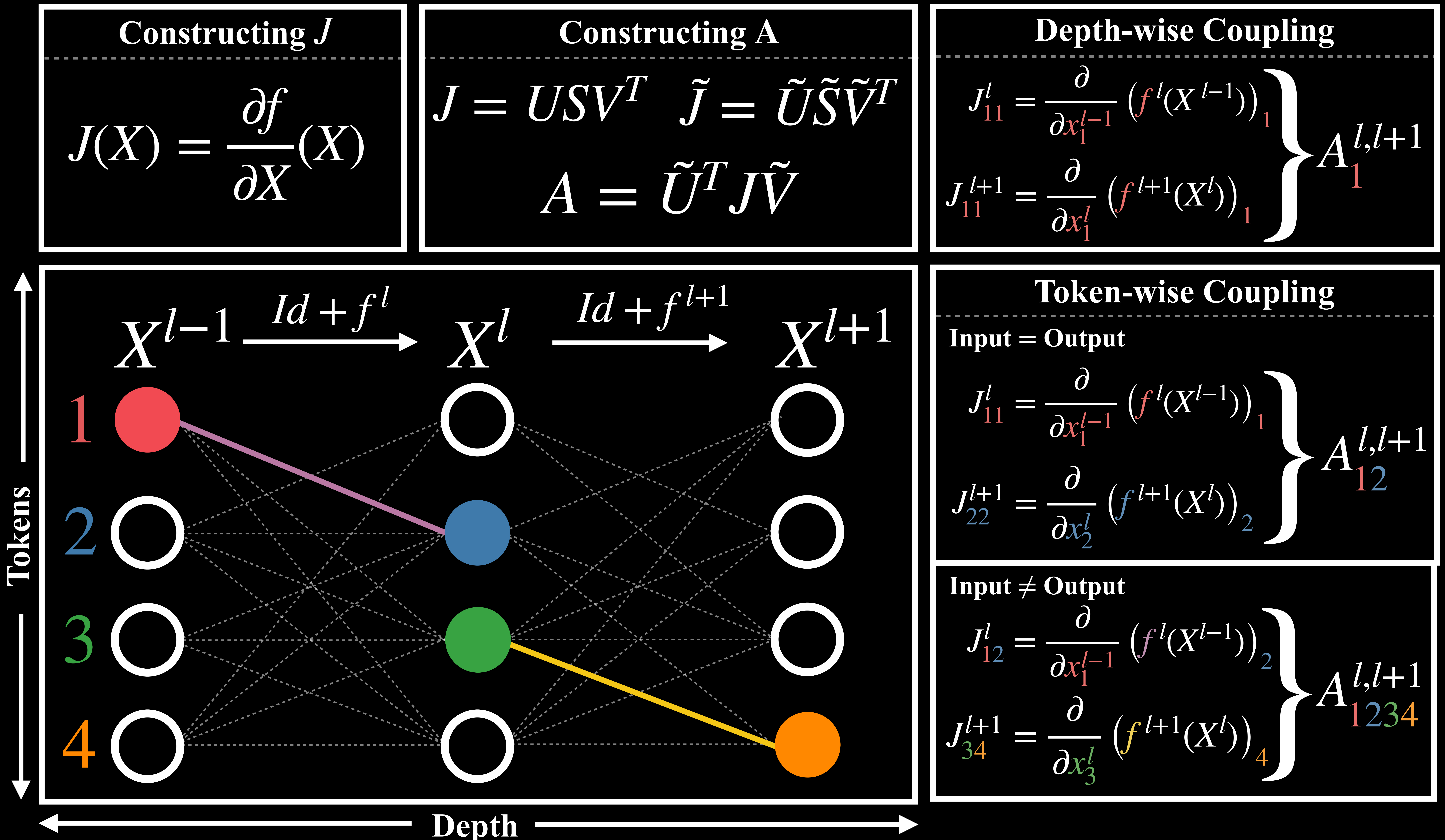
f

12 x_1^{l-1} 2

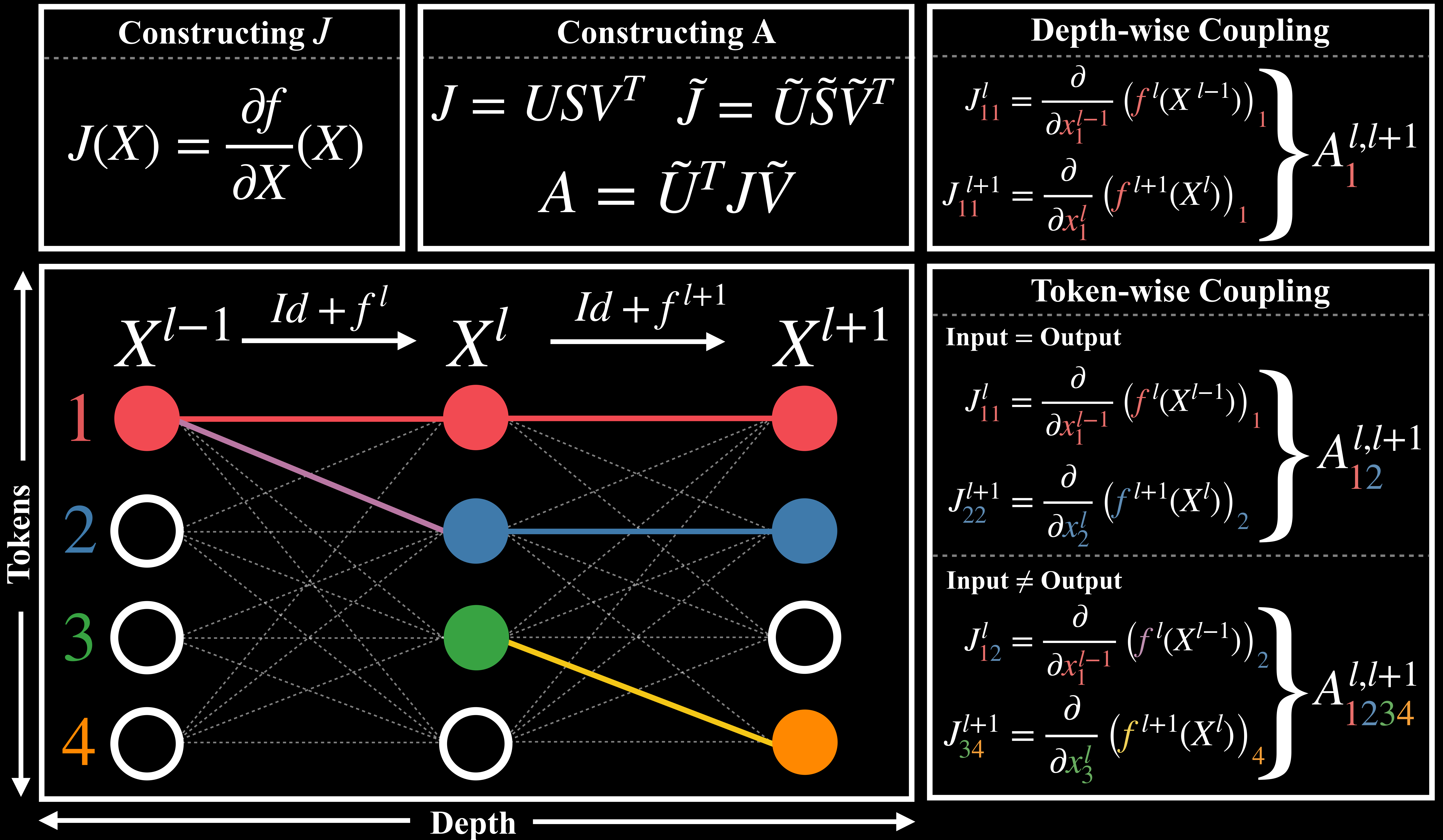
1234

f

34 x_3^{l} 4



Constructing J
$J(X) = \frac{\partial f}{\partial X}(X)$
Constructing A
$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$
$A = \tilde{U}^T J \tilde{V}$
Depth-wise Coupling
$\left.\begin{aligned} J^l_{11} &= \frac{\partial}{\partial x^{l-1}_1}\left(f^l(X^{l-1})\right)_1 \\ J^{l+1}_{11} &= \frac{\partial}{\partial x^l_1}\left(f^{l+1}(X^l)\right)_1 \end{aligned}\right\} A^{l,l+1}_1$
Token-wise Coupling
Input = Output
$\left.\begin{aligned} J^l_{11} &= \frac{\partial}{\partial x^{l-1}_1}\left(f^l(X^{l-1})\right)_1 \\ J^{l+1}_{22} &= \frac{\partial}{\partial x^l_2}\left(f^{l+1}(X^l)\right)_2 \end{aligned}\right\} A^{l,l+1}_{12}$
Input ≠ Output
$\left.\begin{aligned} J^l_{12} &= \frac{\partial}{\partial x^{l-1}_1}\left(f^l(X^{l-1})\right)_2 \\ J^{l+1}_{34} &= \frac{\partial}{\partial x^l_3}\left(f^{l+1}(X^l)\right)_4 \end{aligned}\right\} A^{l,l+1}_{1234}$
$X^{l-1} \xrightarrow{Id + f^l} X^l \xrightarrow{Id + f^{l+1}} X^{l+1}$
1
2
3
4
Tokens
Depth


Constructing J
$J(X) = \frac{\partial f}{\partial X}(X)$
Constructing A
$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$
$A = \tilde{U}^T J \tilde{V}$
Depth-wise Coupling
$J^l_{11} = \frac{\partial}{\partial x^{l-1}_1}\left(f^l(X^{l-1})\right)_1$
$J^{l+1}_{11} = \frac{\partial}{\partial x^l_1}\left(f^{l+1}(X^l)\right)_1$
$A^{l,l+1}_1$
Token-wise Coupling
Input = Output
$J^l_{11} = \frac{\partial}{\partial x^{l-1}_1}\left(f^l(X^{l-1})\right)_1$
$J^{l+1}_{22} = \frac{\partial}{\partial x^l_2}\left(f^{l+1}(X^l)\right)_2$
$A^{l,l+1}_{12}$
Input ≠ Output
$J^l_{12} = \frac{\partial}{\partial x^{l-1}_1}\left(f^l(X^{l-1})\right)_2$
$J^{l+1}_{34} = \frac{\partial}{\partial x^l_3}\left(f^{l+1}(X^l)\right)_4$
$A^{l,l+1}_{1234}$
X^{l-1}
$Id + f^l$
X^l
$Id + f^{l+1}$
X^{l+1}
1
2
3
4
Tokens
Depth